

Nishant Gupta

✉ Email | 📞 +91-9102702999 | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio

SUMMARY

Third-year CS (Data Science) student at Manipal University Jaipur specializing in ML, GenAI, and full-stack development. Engineered production-grade RAG pipelines, CV models, and AI-integrated web platforms — hackathon finalist (2025). Seeking internship to build and deploy intelligent systems at scale.

PROJECTS

VayuDrishti – Air Quality Forecasting App Hackathon 2025

- Engineered spatio-temporal ML pipeline (150k samples) using CPCB, INSAT-3DR, and ERA5 datasets for 6-class AQI prediction.
- Trained XGBoost + Optuna (50 trials), achieving **86.68% accuracy**; best class F1: Unhealthy 93.19%.
- Diagnosed severe class imbalance (Good: 1.34% vs Unhealthy: 57%), reverting strategies that degraded performance by 10.72%.
- Repo: [GitHub](#) — Demo: [YouTube](#)

KrishiMind SustainAI – Agricultural ML Advisory AMD Slingshot 2026

- Trained 3-model pipeline (Random Forest + EfficientNetB0) for yield prediction, price forecasting (R^2 : 0.9635, RMSE: 151.94), and disease detection (28 classes, 96%+ accuracy).
- Deployed FastAPI + Docker backend and Next.js frontend (4 pages); sub-2s inference across 10+ crop types.
- Repo: [GitHub](#) — Live: [Website](#)

Invincible – RAG-Based Study Assistant 2025

- Architected 8-stage RAG pipeline (9 formats) with dual OCR (Tesseract + Gemini Vision fallback) and ChromaDB vector store.
- Hybrid retrieval (dense + keyword, top-10) with MiniLM-L-6-v2 reranking (top-4) and 10-turn per-session memory.
- Citation-grounded generation (Gemini Flash, temp: 0.2) using context-only prompting for hallucination control.
- Repo: [GitHub](#)

Women's Health AI — Fine-Tuned LLM for Medical Q&A 2025

- Fine-tuned **LLaMA 3.2 3B** using **QLoRA (4-bit)** on 24,444 real doctor-patient conversations via Unsloth on Google Colab T4 GPU, achieving final training loss of **0.698**.
- Built domain-specific women's health assistant covering periods, hormonal health, sexual health, PCOS, menopause, and reproductive disorders with judgment-free response generation.
- Deployed interactive **Gradio** web application on HuggingFace Spaces; exported quantized **GGUF** model for local inference via Ollama.
- Repo: [GitHub](#) — Live: [HuggingFace Space](#)

EDUCATION

2023 – 2027	Manipal University Jaipur , Rajasthan, India B.Tech in Computer Science and Engineering (Data Science)
2023	Class 12 – ST Karen's High School, CBSE
2021	Class 10 – ST Karen's High School, CBSE

SKILLS

Languages:	Python, JavaScript, TypeScript, Kotlin, SQL, C++
ML & AI:	PyTorch, scikit-learn, XGBoost, TensorFlow, LangChain, ChromaDB, Gemini API
Data:	pandas, NumPy, Matplotlib, Feature Engineering, Statistical Analysis
Full Stack:	Next.js, React.js, NestJS, FastAPI, Node.js, REST APIs
Android:	Kotlin, Jetpack Compose, MVVM, Room DB, Coroutines
Databases:	PostgreSQL, MongoDB, MySQL, Redis, pgvector
Tools & Cloud:	Git, Docker, Firebase, Azure AI, Streamlit, Linux

CERTIFICATIONS

IBM Machine Learning Professional Certificate
DeepLearning.AI Deep Learning Specialization
Microsoft Azure AI Fundamentals – Generative AI & Computer Vision

All certificates are available on LinkedIn.